

Part 3

Animations for TF3: events, strips and direction-dependent parts

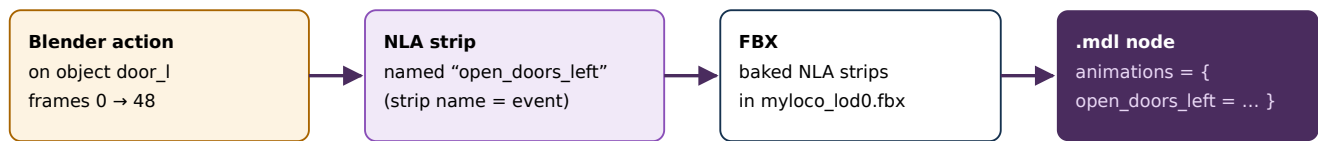
The same keyframes you made for TSC's `.ia` files become TF3 `.ani` files, but they are triggered by game events, not by control values. This part lists the rail-vehicle event vocabulary, how to name strips in Blender, and how pantographs, doors, wheels and lights map.

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Event names are quoted from the Vehicle Types and Vehicle Advanced Topics pages; NLA import behaviour from the Model Editor page.

1 · How an animation gets from Blender into TF3



“In Blender, the animations need to be saved as non linear animations (NLA)... the strips themselves have the correct name of the animation event.”

Figure 3.1 · Strip name = event name manual. One strip per animated object; several objects answering the same event use the same strip name with `#...` to keep Blender from appending `.001`. The add-on pushes tagged actions into strips for you.

2 · The rail vehicle event vocabulary

Event / channel manual	Fired when	Typical node
<code>forever</code>	Always; looped; any length.	Roof fan, radar, beacon
<code>drive</code>	Continuously, mapped to current speed.	Rods, valve gear, anything speed-driven
<code>wheels</code>	Continuously; animation length mapped to one wheel rotation by the engine (rail only).	Wheels that are not plain axle nodes
<code>open_all_doors / close_all_doors</code>	Stop at a station, both sides.	Doors
<code>open_doors_left / close_doors_left , ..._right</code>	Stop with platform on that side.	Doors per side
<code>brake_lights_on / brake_lights_off</code>	Braking starts / ends.	Tail lamps
<code>blink_lights_left_on/off , ..._right_...</code>	Trams only: turning / departing.	Indicators
<code>front_forward_parts_on/off , inner_forward_... , back_forward_...</code>	Position in the consist while driving forward.	Headlights, tail lamps, gangways, pantograph
<code>front_backward_parts_on/off , inner_backward_... , back_backward_...</code>	Same, driving backward (reversible consists).	Same, other end
<code>forward_parts_on/off , backward_parts_on/off</code>	Direction only.	Direction-specific details
<code>has_previous_on/off , has_no_previous_... , has_next_... , has_no_next_...</code>	Whether a vehicle is coupled ahead / behind.	End-of-train devices, gangway bellows

“Each position requires both the on and off animation event to be present” and “an instance of a mesh can't be used for multiple of these purposes. To use the same mesh in several cases, add another instance of it.” [manual](#) The base game ships ready-made on/off animations:

`::/vehicle/shared/ani/front_forward_parts_on.ani / _off.ani`, referenced as `type = "FILE_REF"`.

[manual](#)

3 • Keyframe rules

Rules the Model Editor validates [manual](#)

- The first keyframe must be at time 0 (“has keyframes starting after 0”).
 - An animation needs at least one keyframe; time stamps and transforms must match in number.
 - Position-mapped channels (ship rudder, aircraft control surfaces) use keyframes 0 → 1000 with 500 as the middle; rail vehicles have none of these, but the same “0...1000 maps to a state” idea applies if you write a transformator that scrubs an animation.
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- Durations are real time: a 48-frame door at 24 fps is 2 s; set `arrivalDelay = 2000` to match. [manual](#)
 - Rest pose at frame 0, as in TSC. The LOD-0 FBX is exported at frame 0.
 - Keep the TSC habits: scale applied, origins on hinges, linear keys where the motion is mechanical.
 - Door/lamp “off” animations are usually one keyframe hiding or moving the part; the paired “on” animation shows it. Both must exist.

4 • Wheels and rods

Plain axles: no animation at all

Nodes listed in `railVehicle.config.axles` are rotated about their y axis by the game from the speed and the mesh radius. [manual](#) This covers every TSC `boNNwhNN` wheelset. Do not key them.

Rods, cranks, valve gear

Key one revolution of the driving wheel and the rods together and put the strips under `drive` (speed-mapped) or `wheels` (length mapped to one rotation). [manual](#) Driving wheels that move rods should then *not* also be in `axles`, or they would be rotated twice [check](#).

5 • Doors and the arrival/departure delays

- 1 Model each door leaf as its own object (origin on the hinge or slide axis), named e.g. `door_l1`, `door_r1`.
- 2 Key open and close as **two actions**, or key open 0→N and close as the reverse; export both as strips `open_doors_left` and `close_doors_left` (right side likewise).
- 3 Set `transportVehicle.arrivalDelay` and `departureDelay` in ms to the longest open / close animation. [manual](#)
- 4 Set `loadSpeed` to roughly the number of door lanes, or `-1` for automatic. [manual](#)

TSC doors (`DoorsOpenClose` control, toggled or Lua-driven) need no re-keying: the open action becomes the open event strip, played in reverse for close if you prefer a single action.

6 · Direction- and position-dependent parts

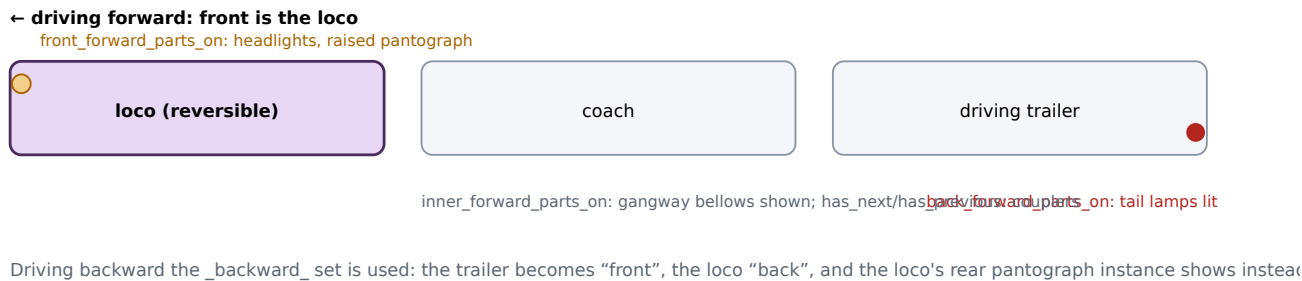


Figure 3.2 · Position × direction. These six on/off pairs and the four `has_next/has_previous` pairs replace most of what TSC vehicles do with `Headlights / Reverser` logic and Lua. [manual](#)

Crew seats follow the same idea: a seat with `forward = true` and `group = "headlights_fwd"` is shown only when that mesh is shown, so the driver appears at the leading end. [manual](#)

7 · Pantographs: the TSC-to-TF3 translation

In TSC a pantograph is a toggled `.ia` on `FrontPantographControl`, raised by the player. TF3 has no player and no pantograph event; the manual's suggestion is “switching pantographs depending on the location in train” with the parts on/off events. [manual](#) Two workable approaches:

A · Mesh swap (no transformator)

Export the pantograph twice: a **raised** instance (keyed at the raised pose) and a **lowered** instance. Give the raised one `back_forward_parts_on/off` (rear pantograph up when the loco is at the front... or whichever convention your prototype used) and the lowered one the opposite pair. Instant, robust, no code.

B · Animated raise (transformator)

Keep the TSC raise animation as a strip named e.g. `panto_raise` and write a transformator that triggers it when the consist starts moving or when the position changes (Part 5). More work; looks better in depots.

For the Texas Electric trolley poles the same two options apply: the pole frame range 0–100 from the TSC spec becomes a single `.ani`; the game-driven “dewire” gameplay has no TF3 counterpart, only the visual (Part 5 §6).

8 · Mapping table: TSC animation → TF3 event

TSC	TF3	Note
Wheels via bogie bp axles	<code>railVehicle.config.axles</code>	No animation needed in either game.

TSC	TF3	Note
Rods (custom keyword nodes, no .ia; TSC animates rods from wheel rotation via linked nodes check)	drive / wheels strips	Key one revolution.
Wipers (BackAndForthContinuous)	forever (looped)	TF3 has no rain trigger for vehicles; wipers run always or not at all check ; most modders omit them.
frontPanto / rearPanto (Toggle)	parts on/off instances, or transformer	\$7.
DoorsOpenClose	open_/close_doors_left/right/all	Set arrival/departure delays.
Headlights + glow meshes	front_/back_forward_parts_on/off + FBX light objects	Base-game on/off .ani can be referenced.
Cab levers, needles	—	No cab in TF3; skip.
Steam smoke emitter child	Emitter locator ...smoke_dark...	Default transformer scales with speed/power.

9 • Troubleshooting

Symptom	Cause → fix
Animation imported but never plays	Strip name is not an event name the rail transformer fires; check spelling against \$2.
“has keyframes starting after 0”	Add a key at frame 0 (rest pose). manual
“incorrect number of times ... and transforms”	Broken FBX bake; re-export with all channels keyed on every frame (“force start/end keying”). manual
Part shows in both directions	Only the _on or only the _off event exists; both are required. manual
Same mesh needed at both ends	Duplicate the object: one instance per purpose. manual
Wheels spin twice as fast	Node is both in axles and keyed under wheels / drive. check
Doors open before the train stops / passengers board through closed doors	arrivalDelay / departureDelay shorter than the animation. manual